

Quantum Communications and Networks(QU.50013)

Exercise Set 1

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Problem 1

The benefits of applying quantum technology come not from communication speeds, but rather the usefulness of sharing quantum states itself. **Quantum key distribution** (QKD) is one of them: to ensure computational security, we use the quantum mechanical principle that perfectly discriminating non-orthogonal states is impossible. Another is **quantum teleportation**, where unknown states can be sent far away, enabling distributed quantum computing systems. In both of these cases, qubits and ebits (entanglement) are the crucial quantum resources.

Problem 2

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